

Christoffel Symbols in Planar Polar Coordinates

Coordinates:

$$(r, \theta)$$

Metric:

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$

Inverse metric:

$$g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}$$

Only nonzero derivative:

$$\frac{\partial}{\partial r}(r^2) = 2r$$

The Christoffel symbols are given by:

$$\Gamma_{ij}^k = \frac{1}{2} \left[g^{k1} \left(\frac{\partial g_{j1}}{\partial x^i} + \frac{\partial g_{i1}}{\partial x^j} - \frac{\partial g_{ij}}{\partial r} \right) + g^{k2} \left(\frac{\partial g_{j2}}{\partial x^i} + \frac{\partial g_{i2}}{\partial x^j} - \frac{\partial g_{ij}}{\partial \theta} \right) \right]$$

Explicit Calculations

1. Γ_{rr}^r

$$\Gamma_{rr}^r = \frac{1}{2} \left[1(0 + 0 - 0) + 0(0 + 0 - 0) \right] = 0$$

2. $\Gamma_{r\theta}^r$

$$\Gamma_{r\theta}^r = \frac{1}{2} \left[1(0 + 0 - 0) + 0(0 + 0 - 0) \right] = 0$$

3. $\Gamma_{\theta r}^r$

$$\Gamma_{\theta r}^r = \frac{1}{2} \left[1(0 + 0 - 0) + 0(0 + 0 - 0) \right] = 0$$

4. $\Gamma_{\theta\theta}^r$

$$\begin{aligned} \Gamma_{\theta\theta}^r &= \frac{1}{2} \left[1 \left(0 + 0 - \frac{\partial}{\partial r}(r^2) \right) + 0(0 + 0 - 0) \right] \\ &= \frac{1}{2}(-2r) = -r \end{aligned}$$

5. Γ_{rr}^θ

$$\Gamma_{rr}^\theta = \frac{1}{2} \left[0(0 + 0 - 0) + \frac{1}{r^2}(0 + 0 - 0) \right] = 0$$

6. $\Gamma_{r\theta}^\theta$

$$\begin{aligned}\Gamma_{r\theta}^\theta &= \frac{1}{2} \left[0(0+0-0) + \frac{1}{r^2} \left(\frac{\partial}{\partial r}(r^2) + 0 - 0 \right) \right] \\ &= \frac{1}{2} \cdot \frac{2r}{r^2} = \frac{1}{r}\end{aligned}$$

7. $\Gamma_{\theta r}^\theta$

$$\begin{aligned}\Gamma_{\theta r}^\theta &= \frac{1}{2} \left[0(0+0-0) + \frac{1}{r^2} \left(\frac{\partial}{\partial r}(r^2) + 0 - 0 \right) \right] \\ &= \frac{1}{r}\end{aligned}$$

8. $\Gamma_{\theta\theta}^\theta$

$$\Gamma_{\theta\theta}^\theta = \frac{1}{2} \left[0(0+0-0) + \frac{1}{r^2} (0+0-0) \right] = 0$$

Final Result

The only nonzero Christoffel symbols are:

$$\begin{aligned}\Gamma_{\theta\theta}^r &= -r \\ \Gamma_{r\theta}^\theta &= \Gamma_{\theta r}^\theta = \frac{1}{r}\end{aligned}$$